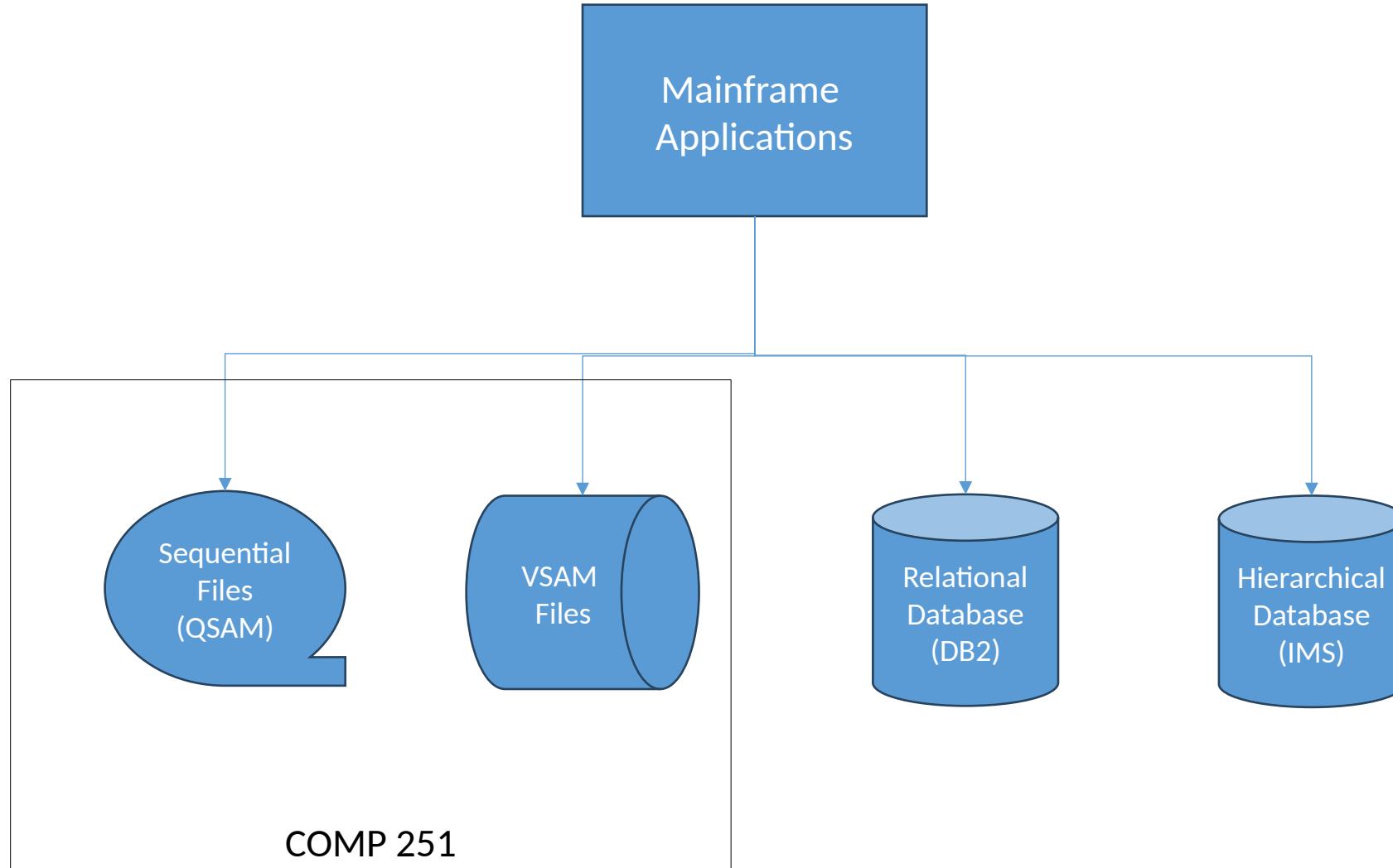


File Organization and Access Mode

COMP251 F2025

Mainframe Data Storage



File Organization

In COBOL, file organization refers to the method used to store and arrange data records within a file.

COBOL supports three types of **file organization**

- Sequential file
- Relative file
- Indexed file

Sequential File

- Records are stored one after another in the order they are written (entry order).
- To read a record, all preceding records must be read first.
- Records do not need to be ordered by a key field.
- Updating or deleting records requires rewriting the entire file
- The access mode allowed in a program is sequential only.

Relative File

- Records are identified by their location relative to the beginning of the file, e.g.,
 - 1st record has a relative record number of 1
 - 10th record has a relative record number of 10, and so on.
- Access modes in a program can be sequential, random, or dynamic.
- When relative files are read or written sequentially, the sequence is that of the relative record number
- Example:
 - if student number is a six digit number and is the key then we need a relative file with room for 999,999 records.
 - Relative files need to be “pre-formatted” , so this student file will be created with 999,999 empty records.
 - When student number 123456 is created, the data is written to record number 123,456.
- Advantage is that I/O is very fast.
- Disadvantage is that it will be a long time before we have one million students, but we are using all that disk space now.

Indexed File

- The file contains a special field whose contents form the record key
- Position of the key is the same in each record
- Has two components: Index and Data
 - Index file establishes the logical arrangement of the file, i.e., in key sequence.
 - Data file is NOT arranged in key sequence. Records are stored in a relative file.
- Can also use alternate indexes, which let you access the file using a different sequence of the records.
- Access modes in a program can be sequential, random, or dynamic

INDEX

123456,3

123674,1

135234,2

DATA

1

123674 ,Bob Smith,123 Main...

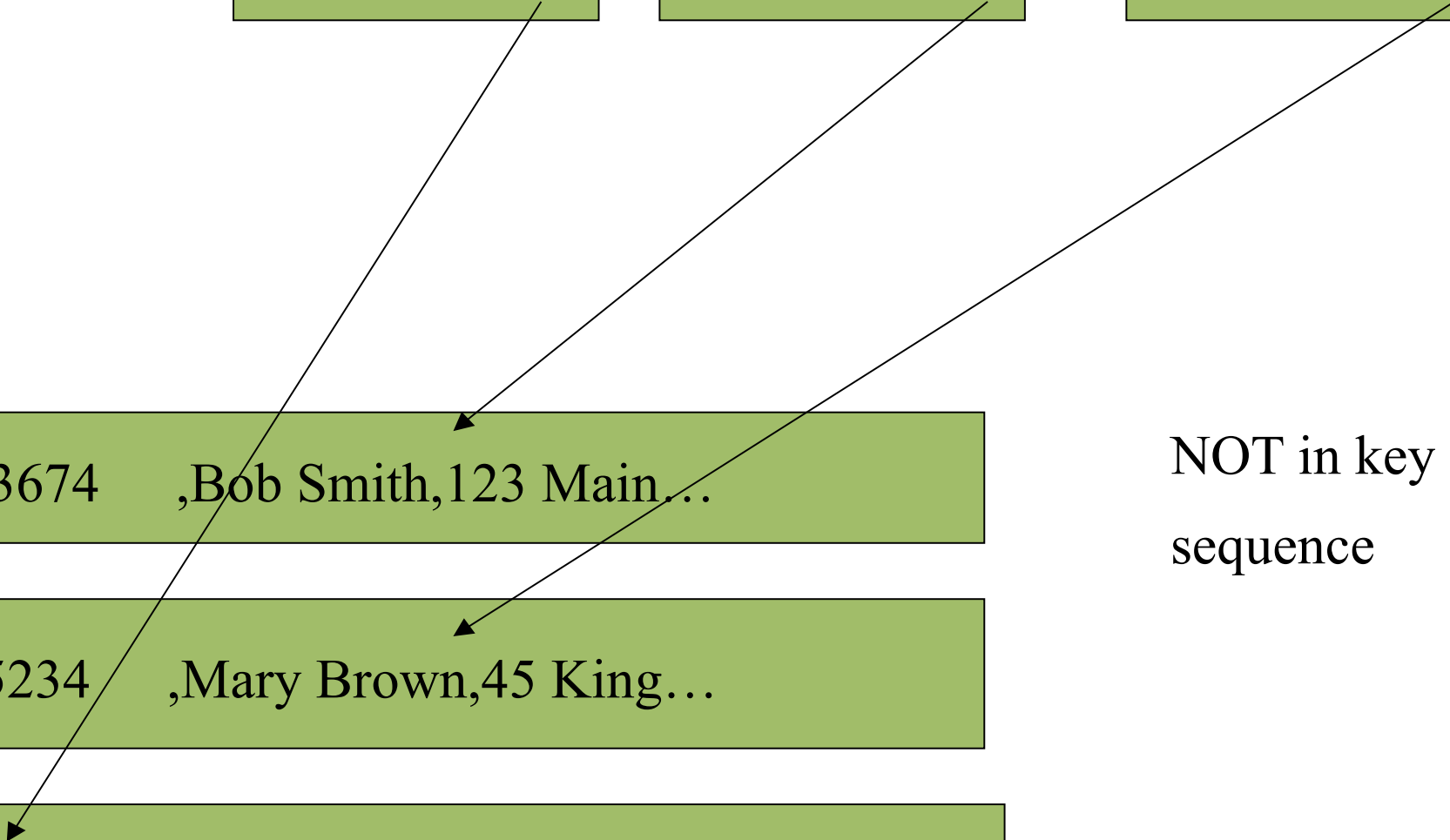
2

135234 ,Mary Brown,45 King...

3

123456 ,Joe Student, 99 Brock...

NOT in key
sequence



What file organization to choose?

- If an application accesses records only sequentially and does not insert records between existing records, a sequential file is the simplest type.
- If an application requires both sequential and random access, an indexed file is the most flexible type.
- Business applications mostly use Sequential File and Indexed File. (Relative Files are rarely used)

Implementation of File Organization in Mainframe z/OS

Mainly two types of files are used for business applications in the industry

1. QSAM files (Queued sequential access method files)

QSAM files are sequential files. It is commonly called a “Flat File” in the industry.

2. VSAM files (Virtual storage access method files)

VSAM Term	COBOL Term
Entry Sequence Data Set (ESDS) <i>(equivalent to QSAM)</i>	Sequential file
Key Sequence Data Set (KSDS)	Indexed file
Relative Record Data Set (RRDS)	Relative file

Access Mode in COBOL

By specifying the access mode in COBOL, you tell the computer how to retrieve records from the file.

COBOL supports three access modes:

1. Sequential access
 - read/write records in serial manner, one record after another.
2. Random access
 - read/write records based on the key value
3. Dynamic access
 - records can be processed sequentially or randomly or both.
 - dynamic access allows you to position to a record randomly and then performing sequential reads, i.e., READ by key and then a few READ NEXT operations.

What access mode to choose?

- If a large percentage of a file is referenced or updated in an application, sequential access is faster than random or dynamic access
- If a small percentage of records is processed during each run of an application, use random or dynamic access.

Relationship between file organizations and access modes

Valid access modes for each file organization

1. Sequential files

- ONLY sequential mode is allowed
 - Sequential mode: records are accessed in the order in which the records were originally written.

2. Indexed files

- All 3 access modes are allowed, i.e., sequential, random and dynamic
 - Sequential mode: records are accessed in ascending order of record key value
 - Random mode: you control the sequence in which records are accessed by specifying the record key value in the **RECORD KEY** data item.
 - Dynamic mode: you can change as needed from random access to sequential access by using appropriate I/O statements, i.e., **READ..KEY IS** and **READ NEXT RECORD**.

3. Relative files

- All 3 access modes are allowed, i.e., sequential, random and dynamic
 - Sequential mode: records are accessed in ascending order of relative record numbers
 - Random mode: you control the sequence in which records are accessed by specifying the relative record number in the **RELATIVE KEY** data item.
 - Dynamic mode: you can change as needed from random access to sequential access by using appropriate I/O statements.